

Tutorial 8: Data Privacy and Data Quality

Question 1

What is a quasi-identifier? Can a dataset have no quasi-identifier, or multiple quasi-identifiers? What is the role a quasi-identifier plays in the k-anonymity approach?

Question 2

There are many data privacy attacking methods, considering the statistical attack example in the course Lecture Notes, how can you protect data privacy (i.e., not revealing individual's salary) using k-anonymity, l-diversity and differential privacy? Please also discuss their potential problems.

Question 3

Explain the following data quality dimensions with examples.

- Integrity
- Accuracy
- Representational consistency
- Completeness
- Timeliness
- Accessibility

What data quality problems, e.g., inaccuracy, inconsistency, duplication, etc. can you find from the following tables?

(a) Student Table

Surname	Firstname	Age	Major	Degree	Sex
Barratt	John	22	Maths	BSc	Male
Burns	Robert	24	CS	BSc	Male
Carter	Laura	20	Physics	MSc	Female
Davies	Michael	12	CS	BSc	Male

(b) Driving Test Table

Surname	Firstname	DoB	Driving test passed
Smith	J.	17/12/85	12/12/05
Smith	Jack	17/12/85	12/12/2005
Smith	Jock	17/12/95	12/12/2005

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